

CS343 Operating Systems

Lecture 3

Introduction to CPU Scheduling



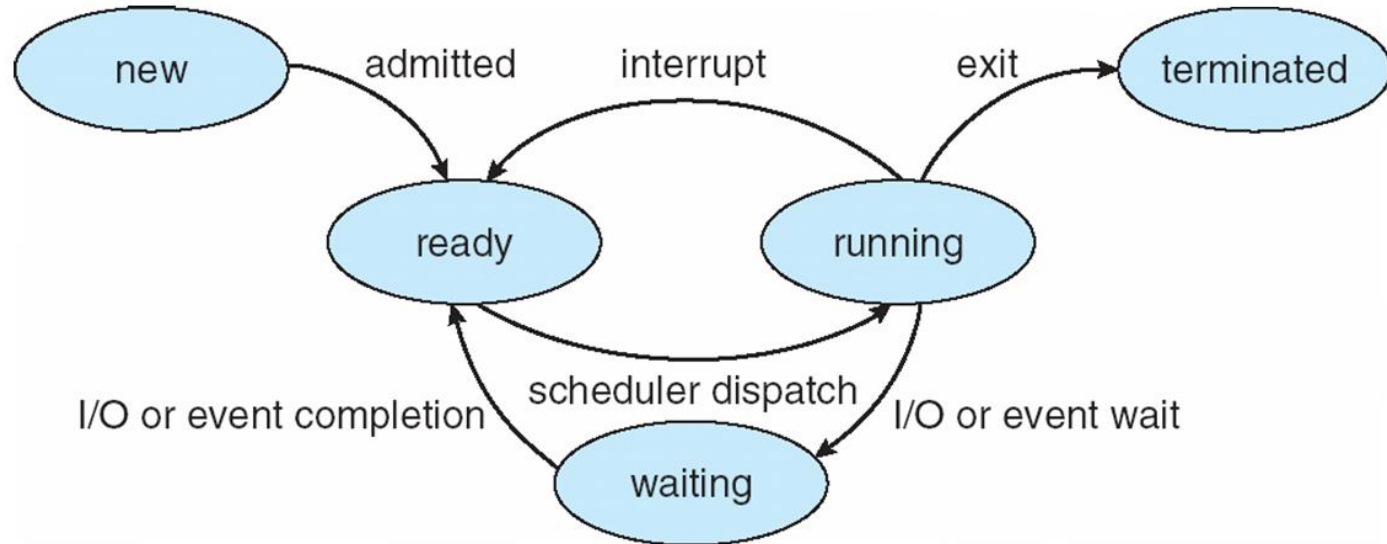
John Jose

Associate Professor

Department of Computer Science & Engineering

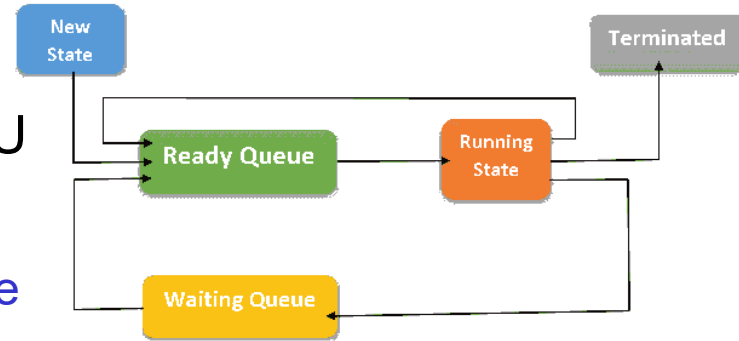
Indian Institute of Technology Guwahati

Process State Diagram



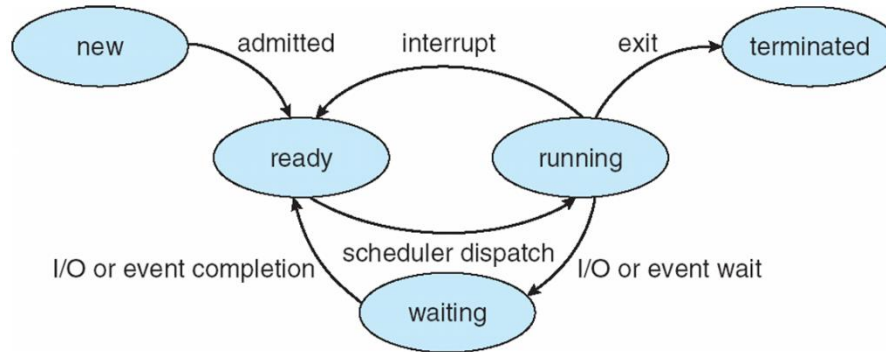
CPU Schedulers

- ❖ **Short-term scheduler (CPU scheduler)** selects from among the processes that are ready to execute and allocates the CPU to one of them.
 - ❖ Selection from **Ready state** to **Running state**
 - ❖ Short-term scheduler is invoked frequently



Preemptive vs Non-preemptive Scheduling

❖ CPU-scheduling decisions happen during the four circumstances:



1. When a process switches from the running state to the waiting state
2. When a process switches from the running state to the ready state
3. When a process switches from the waiting state to the ready state
4. When a process terminates

Preemptive vs Non-preemptive Scheduling

- ❖ CPU-scheduling decisions happen during the four circumstances:
 1. When a process switches from the running state to the waiting state
 2. When a process switches from the running state to the ready state
 3. When a process switches from the waiting state to the ready state
 4. When a process terminates
- ❖ 1 & 4, the scheduling is **non-preemptive** (cooperative)
- ❖ A running process keeps the CPU until it releases the CPU (terminating or by switching to the waiting state).
- ❖ 2 & 3, the scheduling is **pre-emptive**
- ❖ The process is removed from running state forcefully.

Scheduling Criteria

- ❖ Different CPU-scheduling algorithms have different properties.
- ❖ Certain characteristics/criteria are used for comparing various CPU scheduling algorithms.
 - ❖ CPU Utilization
 - ❖ Throughput
 - ❖ Turnaround time
 - ❖ Waiting Time
 - ❖ Response Time

Scheduling Criteria

- ❖ **CPU Utilization** – Percentage time CPU is busy executing process. ↑
- ❖ **Throughput** - Number of processes that are completed per time unit. ↑
- ❖ **Turnaround time** - The interval from the time of submission of a process to the time of completion. ↓
- ❖ **Waiting Time** - Amount of time that a process spends waiting in the ready queue. ↓
- ❖ **Response Time** - Time from the submission of a request until the first response is produced. ↓

Categories of Scheduling Algorithms & Goals

❖ Batch System

- ❖ Complete maximum number of jobs per unit time.
- ❖ Minimize time between submission and termination
- ❖ Keep CPU busy all time

❖ Interactive System

- ❖ Response to requests quickly
- ❖ Reduce waiting time
- ❖ Meet user expectations

❖ Real time System

- ❖ Meeting deadlines
- ❖ Ensure quality constraints

CPU Scheduling Algorithms

Batch Systems

- ❖ First-come first-served
- ❖ Shortest job first
- ❖ Shortest remaining Time next

Interactive Systems

- ❖ Round-robin scheduling
- ❖ Priority scheduling
- ❖ Multiple queues
- ❖ Shortest process next
- ❖ Guaranteed scheduling
- ❖ Lottery scheduling
- ❖ Fair-share scheduling

FCFS Scheduling

- ❖ Simplest CPU-scheduling algorithm
- ❖ First-come, first-served - process that requests the CPU first is allocated the CPU first
- ❖ FCFS policy is managed with a FIFO queue
- ❖ When a process enters the ready queue, its PCB is linked onto the tail of the FIFO queue
- ❖ When the CPU is free, it is allocated to process at the head of the queue
- ❖ The running process is then removed from the queue
- ❖ It is non-preemptive, once scheduled it will complete
- ❖ Short jobs wait for long

FCFS Scheduling

❖ Example: Three processes arrive in order P1, P2, P3 all at time 0.

❖ P1 burst time: 24

❖ P2 burst time: 3

❖ P3 burst time: 9



❖ Waiting Time

❖ P1: 0, P2: 24, P3: 27

❖ Completion Time:

❖ P1: 24, 2: 27, P3: 36

❖ Average Waiting Time: $(0+24+27)/3 = 17$

❖ Average Completion Time: $(24+27+36)/3 = 29$

SJF Scheduling

- ❖ Shortest (in terms of CPU time) job available is scheduled first
- ❖ Shorter processes makes progress
- ❖ SJF policy is managed with a priority queue with burst time as input
- ❖ When a process enters the ready queue, its PCB is linked onto the priority queue at the appropriate entry
- ❖ When the CPU is free, it is allocated to the process at the head of the priority queue
- ❖ If too many short jobs, long processes will starve
- ❖ SJF is non-preemptive; once allotted the process will complete
- ❖ Lowest turnaround time

SJF Scheduling

❖ Consider 3 process P2, P3, P1 all arriving at time T0.

❖ P1 burst time: 24

❖ P2 burst time: 3

❖ P3 burst time: 9



❖ Waiting Time

❖ P1: 12, P2: 0, P3: 3

❖ Completion Time:

❖ P1: 36, P2: 3, P3: 12

❖ Average Waiting Time: $(12+0+3)/3 = 5$ (compared to 17)

❖ Average Completion Time: $(36+3+12)/3 = 17$ (compared to 29)

SRTF Scheduling

- ❖ Shortest Remaining Time First (SRTF) job is scheduled first
- ❖ Preemptive scheduling algorithm
- ❖ A priority queue with remaining time is used as input
- ❖ When a process enters/re-enters the ready queue, its PCB is linked onto the priority queue at the appropriate entry
- ❖ When the CPU is free, it is allocated to the process at the head of the priority queue
- ❖ Newly arriving short process may forcefully preempt currently running process.
- ❖ Longer process may have multiple context switch before completion

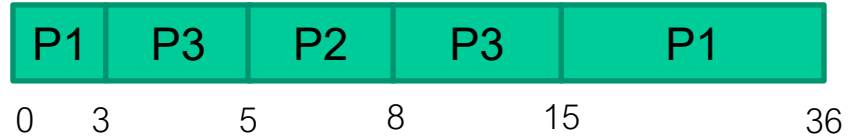
SRTF Scheduling

❖ Consider the following process arriving at different time slots

❖ P1 burst time: 24 arrives at 0

❖ P2 burst time: 3 arrives at 5

❖ P3 burst time: 9 arrives at 3



❖ Waiting Time

❖ P1: $(15-3) = 12$, P2: 0, P3: $(8-5) = 3$

❖ Completion Time:

❖ P1: 36, P2: 8, P3: 15

❖ Average Waiting Time: $(0+3+12)/3 = 5$

❖ Average Completion Time: $(5+10+36)/3 = 17$



johnjose@iitg.ac.in
<http://www.iitg.ac.in/johnjose/>